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An Auction Mechanism for Profit Maximization of Peer-to-Peer Energy Trading in Smart Grids

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Abstract

In this paper, we propose an optimization technique applying an auction-based mechanism for peer-to-peer energy trading in a smart grid. In our proposed system model we have assumed proper infrastructure and communication to allow connected users to trade energy in a peer-to-peer fashion. The mechanism requires complete knowledge of the different constraints tied to the grid infrastructure such as the status of the paths connecting the suppliers and consumers. This view on the underlying infrastructure is imperative not only to prevent power losses and over utilization of the power lines but also to factor the cost of energy transfer in the trading process. We investigate a sealed bid auction model where each player can participate in the bidding process to maximize its benefit according to current and future energy availability, price variations, the distance between traders and energy transfer costs. The evaluation of the proposed model is examined through theoretical analysis as well as experimental results. This work further demonstrates a profit maximization algorithm (PMA) for energy suppliers which generates high profit return.

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Keywords: Peer-to-Peer energy trading; smart grids; profit maximization; auction; optimization

1. Introduction

Recently, there has been wide interest for optimization techniques in peer-to-peer energy trading in smart grids. The smart grid uniquely combines communication and a variety of renewable energy sources readily available for consumers and producers to trade in a peer-to-peer fashion [6]. It provides the infrastructure to allow renewable energy sources to flow among users such as homeowners, commercial and industrial business, etc. This has been shown in [3], where type III-V semiconductors for photovoltaic applications have reached efficiencies close to 30%. Peer-to-peer energy trading lets consumers purchase and sell electricity to one another, allowing consumers to outsource their

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electricity independent of the power grid. Within this paradigm a user can be consumer or producer. For example, a user whose generation of renewable energy exceeds local use can be sold directly to a neighbour with a plug-in EV at a price that suits both. This autonomous choice on purchase and sale of energy allows users to maximize their monetary benefits through optimization for the most favorable energy transactions at any given time. The main problem that faces users is the development of mechanisms that can truthfully maximize their payoff. This is challenging because the price of electricity varies over time based on supply and demand as well as the cost of using the transmission lines. The main goal for the user participating in energy trading is reduce the total energy expense or generate a profit. Several studies such as those in [14], [17], [16] and [4] to name few, suggested market models for grid control, electricity trading and communication. These three properties are the core components for grid functionality.

In this paper, we propose an optimization model to maximize the profit generated from energy producers by selling their energy in a peer-to-peer fashion. The proposed model considers the transmission costs, buyers bid offering, and operating costs. We apply a sealed-bid auction that allows all buyers to fairly present their bid offer. We evaluated the proposed model in two scenarios. In the first scenario, we test when the grid is in non peak-time and in the second scenario we test the grid during peak time. We compared the profit during peak time and non-peak time. When the auctioning process occurs we consider all demands presented from the buyers. The results have shown that our proposed model generates a substantial profit for the supplier and satisfies wider range of buyers.

This paper is organized as follows: Section 2 reviews and discusses related works regarding smart grid applications. In Section 3 the system model is proposed. In Section 4 the optimization method is presented with the main objective function and the constraints. Section 5 discusses the trading mechanism and the algorithm is presented. In Section 6 a performance evaluation is demonstrated and the simulation results are shown. Lastly, in Section 7 we conclude this work and discuss future work.

2. Related Works

The concept of peer-to-peer energy trading among energy suppliers anf buyers have attracted extensive attention from industry and researchers. Several energy trading mechanisms are proposed in the open literature. For example, the work in [1] implements energy trading priorities for users connected to a residential micro grid. Moreover, the work allows individual buyers and sellers to trade energy among themselves before trading with the main grid. This was to address issues nearby homes may experience such as power outages. The study proposes a game-theoretic approach to minimize the overall electricity bills of participants. An alternative approach to address the incentive for income was demonstrated in [11]. In which the authors disregard the profit motivation and consider the value of alternative energy sources. This is done by treating energy sources as heterogeneous products. By allowing individuals to choose by personal preferences based on the inherent attributes of the energy source.

The work in [9] implements a technique for a day ahead approach for virtual power plant owners. This allowed for accurate predications by taking into account weather forecasts, which causes fluctuation in energy generation for solar and wind producers. Thus, looking to optimize the power plant owners profit income. In [5], the authors analyzed the effects that plug in vehicles cause to decentralized grids. The effects arise from the erratic behaviour the peak demand experiences and congestion in network capability. They further demonstrated how these storage systems can be beneficial to the grid by providing alternative storage sources. This relieves balancing and congestion issues for the main grid. Since in some practical situations the main grid itself will experience limitations other may not. These types of grids mainly arise in isolated communities. Where the energy production of the grid itself may be of major concern.

These measures were considered in [8]. The authors took into account that overall energy independence of a micro grid. Where the overall structure of the grid was on an island. This proposed method allows for a useful metric to compare energy trading schemes. In [18] this paper the author used A two-stage Stackelberg game with different cases where Energy users and SFA trade energy. The main motivation of this paper is to encourage all the energy user to participate in energy trading. Here [12] the author presented a block chain method for individual buyer to buy their neighbours excess photovoltaic generated power and in [7] The author Shows a similar network with auction mechanism aim is maximizing the social welfare

In [4], a distributed and centralized market scheme for an energy trading network was investigated. The goal was to provide an overview and comparison between markets by determining which benefits the local market users

(prosumers and consumers) based on prot returns and further provide a connection between these two markets. The system model's objective was to minimize the energy self-consumption of prosumers. Further, the distributed market, which focuses on a peer-to-peer (P2P) exchange approach by dealing with individual buyers and sellers, was based on the continuous double auction (CDA) method. Where as the centralized market focuses on the overall information of all users within the network and looks to maximize the global welfare of the system. This market was implemented using both a energy trading algorithm identifying the market equilibrium and a Vickrey-Clarke-Groves (VCG) method. Based on the results, the local markets performance was found that regardless of the market scheme, both techniques provide benefits to the users. In [15] the author tested the performance of the local markets using continuous double auction mechanism.

In [13] the author discusses cheating in a sealed bid auction by inserting fake bids. Alternatively, in a closed sealed bid auction players cannot cheat. Since buyers are not communicating with each other. The bidding behaviour of power supply consumers under cost assumptions is studied in [19].

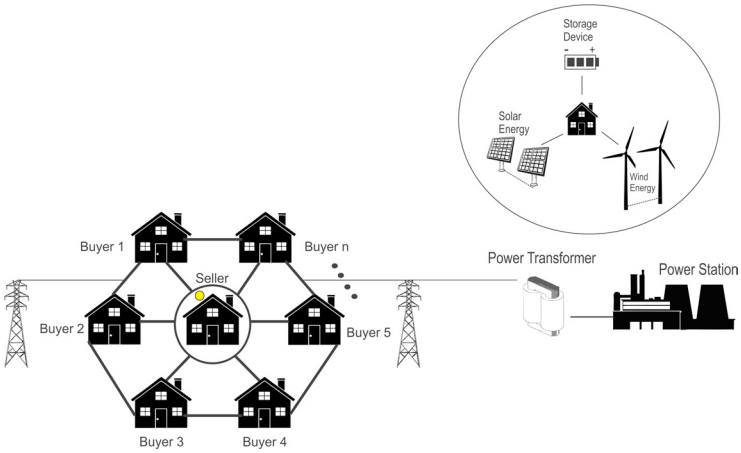

Figure 1: Smart grid network with peer-to-peer energy trading.

3. System Model

Figure (1) provides a general overview of consumers connected to the power grid. In this figure, any house that is capable of acting as an energy supplier or consumer could participate in the peer-to-peer trading. We consider a peer-to-peer energy network consists of a set of suppliers $S = \{1, 2, 3..., n\}$ who produce energy and a set of buyers $B = \{1, 2, 3..., m\}$, where $i \in S$ and $j \in B$. We assume that the supplier has a capacity of generating energy given as E^i and the total demand of the buyers is given as W^j. The set of demands from the buyer is given as $\alpha = \{d_1, d_2, d_3, ..., d_m\}$, $d_j \in \alpha$ and b_m is the demand from the last buyer. The set of bid from the buyers to the supplier is denoted by $\beta = \{b_1, b_2, b_3, ..., b_k\}$, $b_j \in \beta$ and k is the maximum number of bids. We also consider that the electricity grid as one of the buyers during peak time. The operating cost, C_i is the cost for the supplier to produce and maintain energy, E^i. The peer-to-peer power network consists of several possible paths from the suppliers to the buyers denoted as $P_{(i,j)} = \{P^1_{(i,j)}, P^2_{(i,j)}, P^3_{(i,j)}, ..., P^z_{(i,j)}\}$, where z is the maximum number of paths between an energy supplier i and a buyer j. And, each path has its own line cost and transmission capacity, denoted as C^p_l and X^p_{Cap} respectively. The supplier's cost consists of transmission line cost C^p_l and energy generating cost C^p_i. The energy that already flowing through a path is given as $Y^p_{g(i,j)}$ and the energy distributed by the seller to the buyer as $E_{k_{(i,j)}}$. Therefore, the transmission cost is the product of the distributed energy ($E_{k_{(i,j)}}$) and the line costs in the path (C^p_l).

When the transmission cost C^p_l and the operating cost C^p_i are greater than the buyers bid cost b_j, we will assume the supplier stores its energy in a storage bank. The amount of energy stored is denoted by E_y. Where E_y is the difference between the total generated energy E^i and the distributed energy $E_{k_{(i,j)}}$.

4. Problem Formulation

In our model, the suppliers generate renewable energy for local consumption and they can sell the excess of energy to buyers connected to the smart grid. The suppliers objective is to maximize their profit, p_r based on the buyers bid offering. The profit is expressed in equation (1).

$$\mathrm{P_r(i)} = \sum_{j=1}^{m} b_j - \left[C_i + \sum_{P=1}^{z} [E^p_{k_{(i,j)}} \times C^p_l]\right] \tag{1}$$

$$\textbf{s.t.} \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{p=1}^{z} E^p_{k_{(i,j)}} < (X^p_{Cap} - Y^p_{g_{(i,j)}}) \tag{2}$$

$$\sum_{j=1}^{m}\sum_{p=1}^{z} E^p_{k_{(i,j)}} \leq E^i, \forall i \in S \tag{3}$$

The line cost C^p_l in equation (1) is imposed by the electric grid company for using the infrastructure to transmit energy and it varies according to the amount of energy transfered by the supplier. The energy delivered is referred to, as the selling energy, $E^p_{k_{(i,j)}}$. In Equation (2) we are ensuring that the selling energy, $E^p_{k_{(i,j)}}$, never exceeds the transmission line capacity, X^p_{Cap}. This ensures that no path in the system is over-utilized with electric power. To ensure the safety of the transmission grid is not overloaded, the selling energy, $E^p_{k_{(i,j)}}$, should always be less than the difference between the total amount of transmission line capacity and the current amount of electricity flowing in the line, $Y^p_{g_{(i,j)}}$. This relationship is expressed by the following formula, $X^p_{Cap} < Y^p_{g_{(i,j)}} + E^p_{k_{(i,j)}}$. In Equation (3) we ensure that the supplier cannot distribute more than the total energy generated energy, E^i.

When the energy is sold to the grid, we assume the supplier does not need to consider the line cost, since the main grid will cover this expense. Therefore, C^p_l is zero. Also, the supplier does not need to consider selecting the cheapest path since the transmission costs will be zero.

For the buyer, the objective is to minimize the cost of bidding as follows:

$$min \sum_{i=1}^{n}\sum_{p=1}^{z} b_j, \forall j \in B, b_j > 0 \tag{4}$$

$$\textbf{s.t.} \sum_{i=1}^{n}\sum_{p=1}^{z} X^p_{(i,j)} \leq W^j, \forall j \in B \tag{5}$$

Where, $X^p_{(i,j)}$ is the amount of energy buyer $j \in B$ is receiving from supplier $i \in S$ using path p. Equation (5) ensures that the transferred energy does not exceed the total demand W^j of each buyer.

5. Mechanism Design

A sealed-bid auction mechanism is applied to solve the above defined optimization problem [10]. In a sealed-bid mechanism the buyer with the highest bid wins but in this design, due to line cost, the buyer with the highest profit wins. Reverse auction will allow the supplier to decide whether to sell or store their energy. Here, in this setting we consider only one seller and multiple buyers. This mechanism is designed for fairness where all the buyers have a chance to present their bid offers. This ensures that buyers with the minimum demand can compete with other buyers that require a large demand of energy. The profit of the supplier depends on the bid from the buyers b_j and the losses due to transmission cost C^p_l and operating cost C^p_i. The supplier cannot change the operating cost but can reduce the transmission cost by selling their energy to a nearby buyer in the network to further maximize their profit. This will increase the payoff of the supplier.

All active members in the network are considered as players. The buyers in our setting do not communicate with each other. They are allowed to only communicate to the supplier regarding their demand and bidding cost for the selling energy. The auction is initialized by a set of buyers, B, with a set of bids, β. Where $b_j = \{b_1, b_2, b_3, ..., b_k\}$ and $b_j \in \beta$ where $b_j > 0$. We assume the buyer have their own evaluation for the amount of energy, $E^p_{k_{(i,j)}}$.

If $b_j < (C^p_i + C^p_l)$ the profit is negative and the supplier should store their surplus energy. The bid β will be from the buyers according to their evaluation on the amount of energy $E^p_{k_{(i.j)}}$. Evaluation of the buyers is given by $v_j = \{v_1, v_2, ..., v_m\}$. If the profit is maximum for a particular buyer j, then the payoff P_f of that particular buyer is the difference of the evaluation v_j and the bid offering b_j. The other buyers $-j$ receive a payoff of 0. For the payoff of the buyer, if the bid offering is equal to the maximum profit P_{rofit}, the buyer will pay the cost equal to the bid offered. Otherwise, the payoff will be zero and they do not receive the energy.

$$P_f = v_j - b_j, \forall j \in B \tag{6}$$

The problem discussed in the above section can be solved using the PMA which is proposed below.

5.1. Algorithm

Input Buyers demand d_j, Buyers bidding cost b_j, Transmission line capacity X^p_{Cap}, Grid flow Y^p_g, Operating cost C^p_i, and Maximum stored energy E^i.
Output Selling energy $E^p_{k_{(i,j)}}$, and Profit $P_r(i)$.

```
Stage:1 Determining the selling energy, E^p_k
foreach d_j ∈ α do
    Sort d_j in ascending order
    if (min(d_j) < E^i) then
        min(d_j) = E^p_k;
    else
        E^i = E^p_k;
    end
end
Stage:2 Determining the winning buyer
foreach b_j ∈ β do
    Calculate P_r(i)
    select the maximum P_r(i), j
    if (P_r(i) > 0) then
        Check constraints for j
        if (E^p_k < (X^p_Cap + Y^p_g)) then
            sell it to the buyer
        else
            eliminate that buyer j
            Repeat for j-1
        end
    else
        store the energy
    end
end
Stage:3 Payment
The buyer with the highest profit pays the b_j to the seller.
Payment = Σ_(j=1)^m b_j
end
```

Algorithm 1: Profit Maximization Algorithm

5.2. Proposed Algorithm

The profit maximization algorithm repeats until the surplus energy of the supplier is sold or the profit becomes negative. In stage 1 the supplier i selects the selling energy according to the minimum demand of the buyer. Where the minimum demand is chosen from the set $\alpha = \{d_1, d_2, d_3, ..., d_m\}$, $b_d \in \alpha$. In stage 2, the buyers in the game who are all interested in the energy $E^p_{k_{(i,j)}}$, will present their bid offers, b_j. The overall goal is to maximize the suppliers profit from selling their surplus energy

Each seller i will calculate their profit from the bid presented by all buyers j in all possible paths as shown in equation (1). From the calculated profit the supplier selects the buyer with the maximum profit and checks all the constraints. Stated in equation (2), $E^p_{k_{(i,j)}}$ should be less than the transmission capacity in all the paths. Equation (3) ensures that the selling energy $E^p_{k_{(i,j)}}$ should always be less than the generated energy, E^i.

If the constraints are not met, the buyer is eliminated and we consider the next buyer with the largest profit. Then the loop is continued until $E^p_{k_{(i,j)}}$ is purchased. Next, the maximum energy stored from the supplier will be changed by the difference of the previously stored total energy and the sold energy. Lastly, the selling energy $E^p_{k_{(i,j)}}$ will be initialized to zero and the loop continues from stage 1.

Finally, the supplier will be left with a small amount of surplus energy which is less than the minimum demand of the buyers. The supplier sets the selling energy to the amount of energy remaining, if the buyer needs that amount of energy they will propose the bidding cost. The loop will be continued from calculating the profit and if the profit is positive the supplier will sell the energy or the excess energy will be stored

Here, the algorithm is designed fair enough to all the buyers to bid their needed energy. That is the main reason the seller is selecting their selling energy, $E^p_{k_{(i,j)}}$, as the minimum demand. The truthfulness is also maintained by all buyers receiving their demand. They are not getting anything more than their demand and reselling it to anyone. A fair algorithm is designed to treat all the buyers $\forall j \in B$ as the same.

6. Performance Evaluation

6.1. Simulation Setting

The algorithm was tested and simulated using C-programming to obtain a solution for the following case studies. We considered two different case studies. For both case 1 and 2, there is a single supplier and 5 buyers. The supplier has 100kWh of surplus energy. The demand from the buyers range from [20-50] kWh. The operating cost of the supplier is 1.2 cents. The assumed transmission capacity is 1000kWh for all possible paths within the network. We considered that the three of the buyers have one path to reach the supplier and the other two buyers have two paths. The average grid flow throughout all possible paths is assumed to be 200kWh. The buyers offering varies from [7-14] cents/kWh [2] in each iteration. It is considered that the main electric grid is a supplier and selling at a high cost compared to individual suppliers. The ranges set are for simulation purposes only. We initialize the simulation with a set of buyer's demand. In case 1 the seller selects the buyer that offers the highest bid for the selling energy. In case 2 the seller follows the profit maximization algorithm.

6.2. Simulation Results

In case 1, the seller receives bid offerings from all buyers for their surplus energy E^i = 100kWh. The seller then calculates the profit P_r for each buyer using equation (1a). The highest profit calculated was 180 cents for a demand of 50kWh from buyer 5. The seller then distributes this 50kWh surplus energy to buyer 5. Similarity, the seller distributes 20kWh and 30kWh to buyer 1 and 2 respectively. All surplus energy was distributed which generated a net profit 330 cents.

In case 2, the supplier selects the minimum demand from the set of buyers $min(d_j)$ = 20kWh. Therefore, the selling energy $E_k(i, j)$ is set to 20kWh. The seller then receives bid offerings b_j for this selling energy and calculates the profit using equation (1a) for each buyer. Next, the seller distributes the energy to the buyer with the highest calculated profit if the constraints are met. In the first iteration the highest profit calculated was from buyer 5 with a value of 250 cents

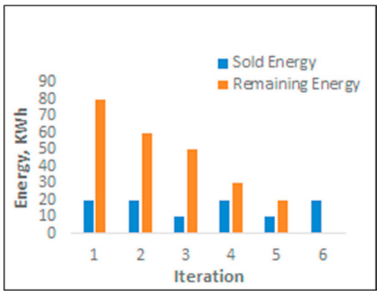

Figure 2: Sold and remaining energy for each iteration from case 2.

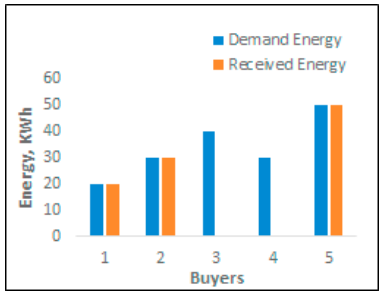

Figure 3: Demand and receieved energy for each buyer.

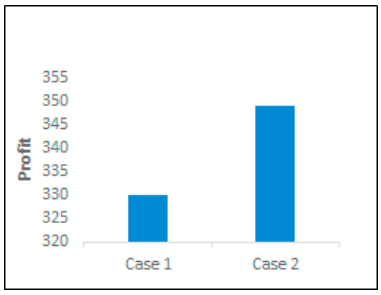

Figure 4: Comparing the profits from the seller in case 1 and case 2

and the constraints were met. Thus, the 20kWh was distributed to buyer 5. Similarly in each iteration the seller selects the minimum demand as the selling energy and repeats the same procedure until all surplus energy is distributed or the profit result is negative. The simulation took six iterations to distribute the 100kWh of surplus energy generated a total net profit of 350 cents.

Fig. 2 represents the sold and remaining energy of the supplier in case 2. In the each iteration the supplier sells 20kWh, 20kWh, 10kWh, 20kWh, 10kWh and 20kWh respectively. Fig. 3 represents the demand of each buyer and how much energy each buyer received from the supplier.

6.3. Comparison

We now make a comparison between case 1 and case 2. Where in case 1 the supplier selects the maximum bid without initializing their selling energy as the minimum demand presented from the set buyers, α. In the proposed algorithm the supplier selects the minimum demand and assigns it to their selling energy, $E^{p}_{k_{(i,j)}}$. In our proposed system model, the iteration count takes longer to compute the optimal buyer but generates a larger profit. Fig. 4 displays the difference in the profits of the supplier between case 1 and the proposed algorithm of case 2. Overall, the profit from our proposed algorithm in case 2 significantly out performs the case 1.

Fig. 5 displays the profit generated during 24 hours for both case 1 and case 2 with an increased amount of buyers and sellers in the network. Each time slot corresponds to a two hour period in energy selling. The profit maxed at t9 which corresponds to peak time during the day where as t3 corresponds to a reduced demand. During peak time slots the grid is a buyer and we can assume suppliers can achieve higher profit returns by selling their generated energy to the grid directly. The results overall demonstrate that the profit is always larger using the profit maximization algorithm (PMA) compared to case 1. It is further demonstrated that the optimal time to sell energy is during peak time.

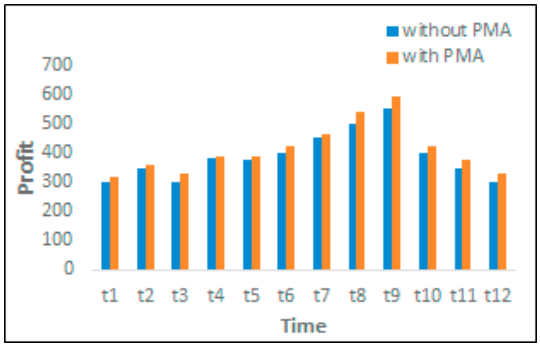

Figure 5: Profit generated by multiple suppliers for different time slots in a day.

7. Conclusion and Future Work

In this paper, we demonstrated an optimization method for peer-to-peer energy trading in smart grids. We consider transmission costs, buyer's bid offer, and operating costs. We allow suppliers to present their surplus of energy to buyers while considering all possible transmission costs associated with each buyer who made a bid and then calculate all possible profits. In our system model, we also ensure that transmission capacities are not exceeded, and that each buyer's demand is met. Based on our simulation results, we applied the auction algorithm for two scenarios, demonstrating how an energy supplier can maximize their profit in different settings. The mechanism is designed to be fair among all buyers in the network. For future work we are planning to extend this work to a comprehensive two-sided bidding auction mechanism where multiple buyers and sellers enter the market for peer-to-peer energy trading. In such setting, we plan to study properties such as truthfulness, individual rationality and computational efficiency in various scenarios of peer-to-peer energy trading.

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